Enhancing Protein Crystallization Prediction through Protein Embeddings

Jayanth Kumar [1] Kavya Jayakumar [2] Jayaprakash Sundararaj [1]

[1]Indian Institute of Technology Bombay [2]Tata Institute of Fundamental Research

Abstract

Predicting protein crystallization is a pivotal task in structural biology, aiding in determining protein three-dimensional structures, essential for elucidating their functions and interactions. This research examines the effectiveness of protein embeddings from two advanced models, ESMCrystal_t6_8M_v1 and ESMCrystal_t12_35M_v2, which are developed based on Meta ESMFold2 [5, 4] architecture. Using datasets from the DeepCrystal, BCrystal, SP, and TR, we evaluate these models' performance in predicting protein crystallization from sequences. Our findings indicate that ESMFold2-based protein embeddings significantly enhance the accuracy of crystallization predictions. The impressive results from both ESMCrystal_t6_8M_v1 and ESMCrystal_t12_35M_v2 models underscore the potential of protein embeddings as a robust tool in predicting protein crystallization, paving the way for more efficient structural biology research.

Introduction

Understanding protein structures is essential in biochemistry and molecular biology. Traditional methods like X-ray crystallography are inefficient. ESMCrystal leverages deep learning and protein embeddings to predict crystallization outcomes efficiently and accurately.

Data and Methods

Datasets Used

Our experiments are performed on publicly available datasets from BCrystal [1] and DeepCrystal [2]. We also, use two independent datasets, SP final (SwissProt) and TR final (Tremble) from [6] for further validation and comparison.

Table 1. Details of Datasets Used in the Study

Dataset	Total Sequences	Crystallizable	Non-crystallizable
DeepCrystal Train	26281	4420	22401
DeepCrystal Test	1898	898	1000
Balanced Test	1787	891	896
SP Test	237	148	89
TR Test	1012	374	638

Training Procedure

Utilized transfer learning from pre-trained ESMFold2 embeddings.

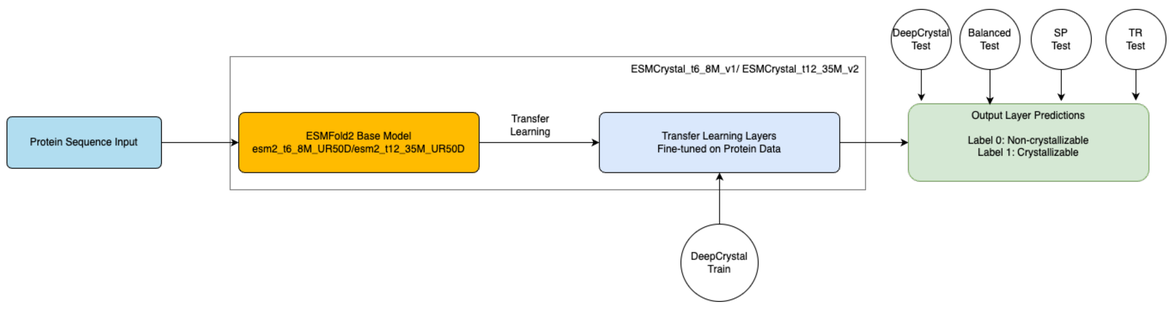

Figure 1. Transfer Learning Architecture for Protein Crystallization Prediction

Models

1. **ESMCrystal_t6_8M_v1**: This smaller model is finetuned on the esm2_t6_8M_UR50D data, with 6 hidden layers and 8 million parameters. The model size is approximately 31.4MB. It is likely faster to train and requires less computational resources but may capture less complex patterns compared to the larger model.
2. **ESMCrystal_t12_35M_v2**: This larger and more complex model is fine-tuned on esm2_t12_35M_UR50D, featuring 12 hidden layers and 35 million parameters, with a total size of approximately 136MB. Since it has more hidden layers, it can capture complex patterns in the protein sequence.

Results

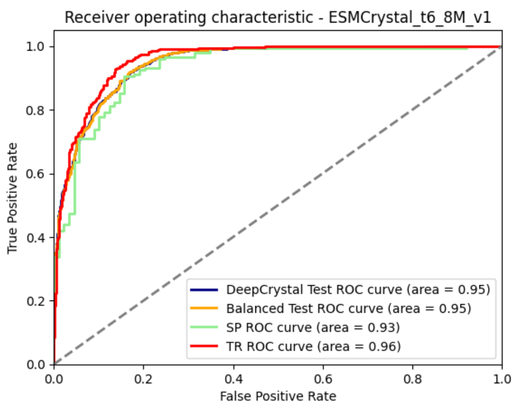

Figure 2. ROC curve for ESMCrystal_t6_8M_v1

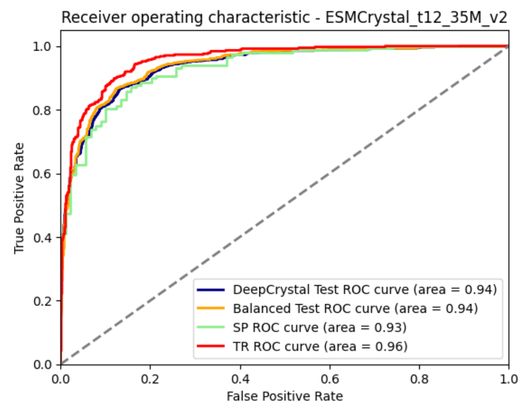

Figure 3. ROC curve for ESMCrystal_t12_35M_v2

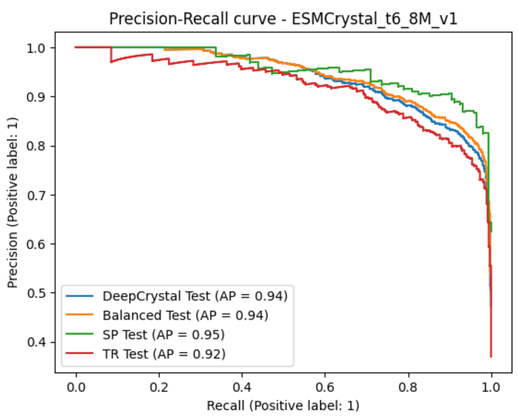

Figure 4. PR curve for ESMCrystal_t6_8M_v1

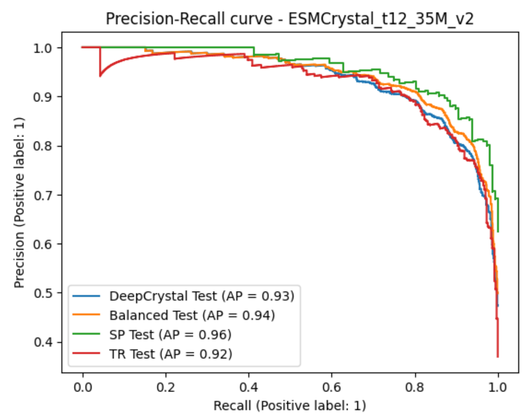

Figure 5. PR curve for ESMCrystal_t12_35M_v2

Results

Both models demonstrated high accuracy, with ESMCrystal_t12_35M_v2 performing better.

Table 2. Performance of ESMCrystal_t6_8M_v1

Dataset	TP	FP	FN	TN	Precision	Recall	F1	Acc	ROC	MCC	NPV
DeepCrystal	532	362	34	966	0.5951	0.9399	0.7288	0.7909	0.9467	0.6119	0.9660
Balanced	531	360	31	865	0.5960	0.9448	0.7309	0.7812	0.9396	0.6045	0.9654
SP	80	68	4	85	0.5405	0.9524	0.6897	0.6962	0.9328	0.5017	0.9551
TR	207	167	16	622	0.5535	0.9283	0.6935	0.8192	0.9562	0.6153	0.9749

Table 3. Performance of ESMCrystal_t12_35M_v2

Dataset	TP	FP	FN	TN	Precision	Recall	F1	Acc	ROC	MCC	NPV
DeepCrystal	579	319	30	970	0.6448	0.9507	0.7684	0.8161	0.9403	0.6575	0.9700
Balanced	573	318	30	866	0.6431	0.9502	0.7671	0.8053	0.9396	0.6446	0.9665
SP Test	97	51	5	84	0.6554	0.9510	0.7760	0.7637	0.9293	0.5861	0.9438
TR Test	225	149	14	624	0.6016	0.9414	0.7341	0.8389	0.9562	0.6588	0.9781

Comparative Analysis

The performance of ESMCrystal models was superior to traditional and other machine learning-based predictors, showcasing the effectiveness of using advanced protein embeddings in this domain.

Table 4. Comparision of ESMCrystal models on Balanced dataset

Models	Accuracy	MCC	AUC	F-score	Recall	Precision	NPV
PPCpred	0.672	0.359	0.754	0.616	0.528	0.740	0.635
fDETECT	0.646	0.355	0.778	0.504	0.360	0.840	0.593
Crysalis I	0.777	0.556	0.865	0.767	0.738	0.799	0.758
Crysalis II	0.804	0.610	0.888	0.796	0.767	0.828	0.784
XtalPred-RF	0.650	0.301	0.710	0.654	0.663	0.645	0.655
TargetCrys	0.627	0.255	0.637	0.637	0.656	0.619	0.593
CrystalP2	0.585	0.177	0.608	0.627	0.700	0.568	0.613
DeepCrystal	0.828	0.658	0.903	0.822	0.795	0.851	0.809
BCrystal	0.954	0.908	0.981	0.954	0.970	0.939	0.969
ESMCrystal_t6_8M_v1	0.7812	0.6045	0.9396	0.7309	0.9448	0.5960	0.9654
ESMCrystal_t12_35M_v2	0.8053	0.6446	0.9396	0.7671	0.9502	0.6431	0.9665

Table 5. Comparision of ESMCrystal models on SP dataset

Models	Accuracy	MCC	AUC	F-score	Recall	Precision	NPV
Crysf	0.700	0.426	0.811	0.727	0.641	0.840	0.572
PPCpred	0.666	0.403	0.784	0.675	0.554	0.863	0.535
fDETECT	0.616	0.381	0.837	0.580	0.425	0.913	0.494
Crysalis I	0.725	0.448	0.835	0.763	0.709	0.826	0.609
Crysalis II	0.751	0.505	0.851	0.783	0.722	0.856	0.633
XtalPred-RF	0.451	0.149	0.449	0.548	0.553	0.564	0.288
TargetCrys	0.611	0.223	0.641	0.659	0.601	0.729	0.486
CrystalP2	0.658	0.257	0.696	0.734	0.756	0.713	0.550
DeepCrystal	0.759	0.530	0.874	0.788	0.716	0.876	0.637
BCrystal	0.894	0.774	0.951	0.919	0.966	0.877	0.932
ESMCrystal_t6_8M_v1	0.6962	0.5017	0.9328	0.6897	0.9524	0.5405	0.9551
ESMCrystal_t12_35M_v2	0.7637	0.5861	0.9293	0.7760	0.9510	0.6554	0.9438

Table 6. Comparision of ESMCrystal models on TR dataset

Models	Accuracy	MCC	AUC	F-score	Recall	Precision	NPV
Crysf	0.841	0.663	0.887	0.747	0.631	0.918	0.817
PPCpred	0.748	0.448	0.819	0.640	0.606	0.677	0.782
fDETECT	0.750	0.447	0.847	0.548	0.411	0.823	0.733
Crysalis I	0.787	0.546	0.870	0.715	0.724	0.707	0.836
Crysalis II	0.816	0.603	0.892	0.748	0.740	0.756	0.849
XtalPred-RF	0.451	0.040	0.525	0.452	0.537	0.390	0.651
TargetCrys	0.634	0.325	0.693	0.614	0.788	0.503	0.733
CrystalP2	0.581	0.241	0.673	0.577	0.775	0.460	0.780
DeepCrystal	0.841	0.657	0.910	0.781	0.762	0.800	0.864
BCrystal	0.963	0.922	0.988	0.951	0.970	0.933	0.982
ESMCrystal_t6_8M_v1	0.8192	0.6153	0.9562	0.6935	0.9283	0.5535	0.9749
ESMCrystal_t12_35M_v2	0.8389	0.6588	0.9562	0.7341	0.9414	0.6016	0.9781

Conclusion

ESMCrystal models represent a significant advancement in protein crystallization prediction, offering a robust, efficient alternative to conventional methods and setting a new standard in applying transfer learning in structural biology. Future refinements could integrate even more advanced models like AlphaFold [3], boosting predictive accuracy and reliability.

References

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 jayanth8506@iitbombay.org